MILITARY SPECIFICATION

VALVE, FUEL INTERLOCK

This Specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

10 SCOPE

1.1 Scope. This specification covers one type of pressure-controlled piston valve referenced herein as the "valve".

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military	
MIL-P-116	Preservation, Methods of
MIL-G-4343	Grease, Pneumatic System

STANDARDS

Military	
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS-16998	Screw, Cap, Socket Head Hexagon Alloy Steel, Cadmium Plated UNF-3A
MS-124735	Insert-Cres Helical Coil Fine Thread, 2 dia. Nominal Length

DRAWINGS

Bureau of Naval Weapons	
DL-2526753	Interlock Valve Assembly, Fuel
2526527	Disc
CD-2526526	Spring
CD-2526566	Body Valve
CD-2526567	Piston
CD-2526608	Cap
CD-2526615	Packing, Preformed

(copies of specifications, standards, drawings, and publications required by contractors in connection with specific procuring functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

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3.1 Classification of requirements. The requirements for the valve are classified as follows:

REQUIREMENT	PARAGRAPH
General Requirements	3.2
Performance Requirements and Product Characteristics	3.3
Environmental Requirements	3.4
Operational Service Life	3.5

3.2 General requirements.

3.2.1 Data requirements. No data is required by this specification or approval documents referenced in Section 2 unless specified in contract or purchase order. See 6.2

3.2.2 Preproduction sample. Unless otherwise specified in the contract or purchase order, the preproduction sample of the valve shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining that prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.3 Materials. Unless otherwise specified in the contract or purchase order, all materials and treatments of materials and parts used in the manufacture of the valve shall be in accordance with the applicable documents listed on BuWeps DL 2526753.

3.2.4 Definitions. Definitions of terms appear in Section 6.

3.3 Performance requirements and Product characteristics. Unless specified otherwise, the valves shall satisfy the following performance requirements when tested as specified in Section 4.

3.3.1 Operation. The valve shall be opened by a pressure of 450 pounds per square inch, gage (psig) applied to the pilot port, causing a piston-driven knife completely to sever a disc.

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3.3.2 Valve closure. With the pilot pressure removed the valve shall close. There shall be no leakage between the inlet and outlet ports.

3.3.3 Pilot cylinder leakage. The pilot cylinder shall not leak with a pressure of 4500 psig applied to the pilot port for a period of five minutes.

3.4 Environmental requirements. Unless otherwise specified, the valve shall be capable of conforming to the performance requirements and product characteristics of the specification after being exposed to the environmental conditions specified in this section. For definitions of ready, operating, and non-operating see 6.3.1, 6.3.2, and 6.3.3.

3.4.1 Vibration, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the vibrations defined in Table I. The valve shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45° F to +140° F. Vibration is defined in 6.3.4.

TABLE I

VIBRATION SCHEDULE

Double Amplitude (in. ± 10%) Minimum	Acceleration (g's ± 10%) Minimum	Frequency (cps)
0.2	---	5 - 22
9.-	3.0	22 - 40
0.036	---	40 - 74
---	10	74 - 500

3.4.2 Acceleration, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the accelerations defined in Table II for a duration of 1 minute for each magnitude. Acceleration is defined in 6.3.5.

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TABLE II
ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)
25	Longitudinal axis, Fwd
35	Longitudinal axis, Aft

3.4.3 Shock, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the shock loads defined in Table III . Shock is defined in 6.3.6.

TABLE III
SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	1800 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity.

3.4.4.1 Non-operating.

a. Temperature. The valve shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, environmental temperatures between -45° F and +140° F. Temperature is defined in 6.3.7.

b. Humidity. The valve shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, an environment of at least 95 percent relative humidity at +110° F for a minimum period of 8 hours. Humidity is defined in 6.3.8.

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3.4.4.2 Ready, operating. The valve immediately after being conditioned for 1 hour at any temperature between 0° F and +140° F shall be capable of meeting the performance requirements at any environmental temperature between 0° F and +140° F.

3.5 Operational service life. The valve shall have a service life of 20 duty cycles.

3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the valve shall be classified as follows:

a. Preproduction inspection (Subsection 4.4)
b. Acceptance inspection (Subsection 4 .5)
c. Periodic inspection (Subsection & .6)
d. Examination of first echelon repair parts (Subsection 4.7)

4.3 Sampling.

4.3.1 Preproduction inspection sample. A preproduction sample of 6 valves manufactured in accordance with 3.2 shall be subjected to the preproduction tests listed in Table IV and detailed in 4.4 through 4.4.4.10 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. See 4.4.6. Further production of the valve by the supplier, prior to approval of the preproduction sample shall be at the supplier's risk.

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4.3.2 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.5. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity for approval, the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Examination of first echelon repair parts. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of first echelon repair parts for the valve as specified in 4.7 through 4.7.4.

4.4 PreProduction inspection.

4.4.1 Tests and examinations. The preproduction sample of 6 valves shall be inspected in accordance with Table IV, with 2 valves being tested in accordance with each of the 3 test sequences, A through C.

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TABLE IV

PREPRODUCTION TEST SEQUENCE

Tests	Paragraph Reference	Test Sequence A	B	C
Operational Test	4.4.4.1	1	1	1
Vibration, Ready Condition	4.4.4.2	2		
Acceleration, Ready Condition	4*4.4.3		2	
Shock, Ready Condition	4.4,4.4		3	
Operational Test	4.4.4.1	3	4	
Temperature and Humidity	4.4.4.5	4	5	
Low Temperature, Operating	4.4.4.6	5		
High Temperature, Operating	4.4.4.7		6	
Vibration, packed and Packaged	4.4.4.8			2
Shock, Packed and Packaged	4.4.4.9			3
Operational Test	4.4.4.1			4
Operational Service Life	4.4.4.10	6		

4.4.2 Test conditions. Unless otherwise specified, the valve shall be subjected to the tests in 4.4.1 under the following conditions:

a. Temperature: Room ambient +32° F to +98° F
b. Altitude: Normal ground atmospheric pressure
c. Vibration: None
d. Humidity: Room ambient

4.4.3 Test equipment and set-up. In addition to the items listed in 4.5.3, test equipment is required which provides the following characteristics:

4.4.3.1 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following characteristics:

a. Frequency range, logarithmic sweep: 5 to 500 cycles per second (cps)
b. Test specimen: 3 pounds
c. Acceleration and displacement (ready condition):

5 to 22 cps	0.2 inch
22 to 40 cps	3.0 gravitational units (g's)
40 to 74 cps	0.036 inch
74 to 500 cps	10 g's

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d. Acceleration (packed and packaged):

5 to 60 cps	1 g

e. Automatic cycling the range: 0 to 15 minutes (optional)

f. Temperature chamber range: -45° F to +140° F accuracy to ± 5°F

g. Interior size: 2-1/2inches x 3-1/2 inches. x 5-1/2 inches minimum .

4.4.3.2 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with a 1 pound test specimen.

4.4.3.3 Shock equipment. The shock equipment shall be capable of accepting a 1 pound test specimen and shall provide ranges of:

a. 300 g's for 0.5 millisecond (ins) to 1.5 ms
b. 60 g's for 8.0 ms to 10.0 ms
c. 45 g's for 18.0 ms to 20.0 ms

4.4.3.4 Temperature and humidity conditioning equipment, One or more equipments may be utilized to provide the following characteristics:

a.	Temperature:	-45° F to +140° F; with an accuracy to ± 5° F
b.	Relative humidity:	95 percent minimum at +110° F; with an accuracy to ± 5° F
c.	Interior size:	2-1/2 inches x 3-1/2 inches x 5-1/2 inches minimum

This equipment may be used for both non-operating and ready/operating temperature (and humidity) tests.

4.4.4 Test Procedures.

4.4.4.1 Operational test. The valve shall be subjected to the acceptance tests detailed in 4.5.4.1 and 4.5.4.2.

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4.4.4.2 Vibration, ready condition. The valve, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 1. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 cps to 500 cps. The valve shall then be subjected to resonance and cycling vibration in each axis, in conformance with the times shown in Table V. If more than 4 resonances are encountered on any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -45° F ± 5° F, room temperature and +140° F ± 5° F respectively. For cycling, a logarithmic frequency sweep rate shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations of double amplitudes defined in Table VI.

TABLE V

VIBRATION TEST SCHEDULE

Number of Resonances	0	1	2	3	4
Total Vibration Time at Resonance *		30 min	1 hr	1-1/2 hr	2 hr
Cycling Time	3 hr	2-1/2 hr	2 hr	1-1/2 hr	1 hr

* 30 minutes each resonance

TABLE VI

VIBRATION SCHEDULE

Frequency (cps)	5-22	22-40	40-74	74-500
Double Amplitude (in. ± 10%) Minimum	0.2		0.036	
Acceleration (g's ± 10%) Minimum		3		10 ;

4.4.4.3 Acceleration, ready condition. The valve, in the ready condition, shall be subjected to the accelerations defined in Table VII after being secured to an acceleration equipment (see Figure 1). Acceleration shall be measured at the approximate geometric center of the valve.

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TABLE VII

ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)	Time
25	Longitudinal axis, Fwd	1 minute
35	Longitudinal axis, Aft	1 minute

4.4.4.4 Shock, ready condition. The valve, in the ready condition, shall be attached to a shock test machine (see Figure..l). Shock magnitudes and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the valve shall be performed in accordance with Table VIII in the sequence shown.

TABLE VIII

SHOCK TEST SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction Of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

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4.4.4.5 Temperature and humidity, non-operating condition. Four preproduction valves shall be subjected, in the non-operating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° F +0° -5° F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -45° F +0° -5° F within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours). The specimen shall then be subjected to an environmental temperature of +140° F +5° F -0° F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours. This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110° F + 5° -0° F at a minimum relative humidity of 95 percent for a minimum of 8 hours. Thereafter, 2 valves shall be operated in accordance with 4.4.4.6 and 2 valves shall be operated in accordance with 4.4.4.7. Prior to operation the valves may be wiped dry. Drying by an air blast or oven is not permitted.

4.4.4.6 Low temperature operating. Before testing, the valve, installed in its test chamber in the ready condition, shall be conditioned at a temperature of 0° F ± 5° F for a period of 1 hour. Immediately thereafter the valve shall be operated in accordance with and meet the requirements of 4.5.4.l and 4.5.4.2 at an environmental temperature of +32° F ± 5° F.

4.4.4.7 High temperature operating. Before testing, the valve, installed in its test chamber in the ready condition, shall be conditioned at a temperature of +140° F ± 5° F for a period of 1 hour. Immediately thereafter the valve shall be operated in accordance with and meet the requirements of 4.5.4.1 and 4.5.4.2 at an environmental temperature of +140° F ± 5° F.

4.4.4.8 Vibration. Packed and packaged. The valve, packed and packaged in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 cps. The container shall be fastened, in suitable fashion, to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The specimen shall then be vibrated for 1 hour at the frequency producing maximum transmissibility and at an input acceleration of 1 g. This test shall be applied to each of the three principal orthogonal axes,

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4.4.4.9 Shock, packed and packaged. The valve, packed and packaged in accordance with Section 5, shall be subjected to free fall drop tests. The valve in its container shall be dropped from a height of 36 inches, one drop on each of 4 diagonally opposite corners and flat drops, one each, on the bottom and on one side. The drops shall be made on an impact surface of essentially non-yielding material such as stone or concrete. The complete series of drops shall be performed under each of the following conditions: room temperature, within 10 minutes after the container has been stabilized at -45° F ± 5° F, and within 10 minutes after the container has been stabilized at +140° F ± 5° F.

4.4.4.10 Operational service life. The valve shall be subjected to 17 additional duty cycles in accordance with 4.5.4.1 and 4.5.4.2.

4.4.5 Test data sheets. All test data shall be recorded on a data sheet by the supplier and maintained as part of the inspection records. In addition to the data requirements of 4.5.5, the data sheets shall contain, but are not limited to, the following information:

a. Date _____
b. Interlock valve serial number _____
c. Preproduction test sequence conducted _____
d. Operational tests: See 4.5.5
e. Preproduction tests

Low temperature operating
Pressure _____ psig
Disc severed _____

High temperature operating
Pressure _____ psig
Disc severed _____

4.4.6 Determination of defect or failure. Failure of a valve to successfully pass any of the tests in 4.4 through 4.4.4.10 shall be cause for rejection of the preproduction sample.

4.5 Acceptance inspection. Each valve produced under the contract or purchase order shall be subjected to the following tests and examinations:

4.5.1 Tests and examinations.

a. Disc rupture and valve closure test
b. Pilot cylinder leak test

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4.5.2 Test conditions. Unless otherwise specified, the valve shall be subjected to the acceptance tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmoshperic pressure
c.	Vibration:	None
d.	Humidity:	Room ambient.

4.5.3 Test equipment and set-up.

a.	Hydraulic supply	0 to 4700 psig minimum
b.	Pressure gage	0 to 5000 psig, accuracy ± 1/2 percent full scale
c.	Dry inert gas supplies	0 to 470 psig, 0 to 160 psig

4.5.4 Test procedures.

4.5.4.1 Disc rupture and valve closure test. The valve shall be assembled to the test fixture in accordance with Figure 1. A gas pressure of 100 ± 10 psig shall be applied to the inlet port. A second source of pressure shall be connected to the pilot port. Apply a gas pressure of 450 psig to the pilot port. This pilot port pressure shall cause the piston knife to sever the disc. Remove inlet and pilot port pressures. Apply dry inert gas at a supply pressure of 150 ± 10 psig to the inlet port for 5 minutes minimum with the valve submerged in water. There shall be no evidence of leakage. Disassemble the valve and inspect the disc. Severing of the disc shall be evidenced by the fact that the OD of the piston shaft at the knife end has passed through the disc. The disc material shall remain intergral.

4.5.4.2 Pilot cylinder leak test. Assemble the valve to the fixture without a disc. Apply a hydraulic pressure of 4500 ± 100 psig to the pilot port for 5 minutes minimum. There shall be no evidence of leakage from the inlet port or the outlet port. Reduce the pilot pressure to 0 psig. Post-test servicing and maintenance shall be performed in accordance with 4.5.7.

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4.5.5 Test data sheets. All test data shall be recorded on data sheets by the supplier. Data sheets shall contain, but are not limited to, the following information:

a. Date
b. Interlock valve serial number
c. Tests:

Disc rupture test
Pressure _____ psig
Disc severed _____

Valve closure
Pressure _____ psig
Time _____ minutes

Pilot cylinder
Pressure _____ psig

4.5.6 Determination of defect or failure. Failure of a valve to successfully pass any of the tests in 4.5.4.1 and 4.5.4.2 shall be cause for rejection of that valve.

4.5.7 Post-test servicing and maintenance. The valve shall be disassembled, cleaned and re-assembled.

4.6 Periodic production inspection. Periodic production samples shall be selected in accordance with 4.3.2. The samples shall be tested in accordance with the following plan:

a. The Sample of 1 unit from the first group of 30 units will be called the first Sample and will undergo the tests of Table IX for Sample 1. The sequence of tests will be as shown in Table IX.

b. The Sample of 1 unit from the second and third groups of 30 units will be called the second and third Samples and will undergo the tests of Table IX for Samples 2 and 3 respectively. The sequence of tests will be as shown in Table IX.

c. The next Sample will be selected from the next group of units regardless of whether the shift to Step 2 was made. This sample will be called the fourth sample and will undergo the tests of Table IX for Sample 4, etc.

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TABLE IX

PERIODIC TEST SEQUENCE

Tests	Sample and Test Sequence		
	1, 4, Etc.	2, 5, Etc.	3, 6, Etc.
Vibration, Ready Condition		1*	1*
Acceleration, Ready Condition			2*
Shock, Ready Condition	1*		3*
Temperature and Humidity		2	4
High Temperature Operating	2		5
Low Temperature Operating		3	6
Operational Test	3	4	
Operational Service Life			7

* Visual Check After Test

4.6.1 Test conditions. Unless otherwise specified, the valve shall be subjected to the periodic tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.6.2 Test equipment and set-up. The test equipment and set-up for the periodic tests shall be in accordance with 4.4.3.

4.6.3 Test procedures.

4.6.3.1 Vibration, ready condition. Vibration testing of the valve, in the ready condition, shall be performed in accordance with 4.4.4.2.

4.6.3.2 Acceleration, ready condition. Acceleration testing of the valve, in the ready condition, shall be performed in accordance with 4.4.4.3.

4.6.3.3 Shock, ready condition. Shock testing of the valve, in the ready condition, shall be performed in accordance with 4.4.4.4.

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4.6.3.4 Temprature and humidity, non -operating condition. Temperature and humidity testing of the valve, in the non-operating condition, shall be performed in accordance with 4.4.4.5.

4.6.3.5 High temperature operating. The valve shall be subjected to the high temperature operating test in accordance with 4.4.4.7.

4.6.3.6 Low temperature operating. The valve shall be subjected to the low temperature operating test in accordance with 4.4.4.6.

4.6.3.7 Operational tests. The valve shall be subjected to the operational tests of 4.5.4.1 and 4.5.4.2.

4.6.3.8 Operational service life. The valve shall be subjected to the operational service life tests in accordance with 4.4.4.10.

4.6.4 Test data. Test data shall be recorded in accordance with 4.4.5, as appropriate, and maintained as a part of the inspection record.

4.6.5 Determination of defect or failure. Failure of the valve to successfully pass the tests in 4.6.3.1 through 4.6.3.8 shall be cause for withholding acceptance of all units on hand. Corrective action shall be taken in accordance with 4.3.2.

4.7 Examination of first echelon repair parts.

4.7.1 General.

4.7.1.1 The information presented in this section is applicable to all components assembled into the end product and to first echelon repair parts. The information is divided into the following major divisions:

a. Parts covered by separate CD's
b. Parts which are manufactured in accordance with Federal or Military specifications which contain adequate quality assurance provisions.
c. Minor hardware items

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4.7.1.2 Unless otherwise specified, sampling plans and procedures set forth in MIL-STD-105, inspection Level II shall be used in determination of acceptability of the first echelon repair parts and all components assembled into the end product. An AQL of 1.0 shall be used for major items of inspection, an AQL of 2.5 shall be used for minor items of inspection.

4.7.1.3 The verification of the quality of repair parts is mandatory and shall be performed whether the item is a part of a repair kit or is to be assembled into the end product.

4.7.2 Separate Classification of Defects(CD's). The components and assemblies referenced within this subsection shall be inspected in accordance "with the applicable CD'S. For inspection of repair parts use:

PART	BU WEPS CD
Spring	2526526
Body Valve	2526566
Piston	2526567
Cap	2526608
Packing, Preformed	2526615
Interlock Fuel Valve Assembly	2526753

4.7.3 Components covered by Federal and Military Specifications. The components and parts referenced within this subsection shall be inspected in accordance with the applicable specifications. For inspection of repair parts use:

PART	SPECIFICATION
Lubricant	MIL-L-4343

4.7.4 Miscellaneous hardware. The items referenced within this subsection shall be inspected as to type and quantity.

PART	IDENTIFICATION
Screw, Cap, Sot. Hd. Hex	MS16998-27
insert	MS124735

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. See 6.2. Unit package shall consist of one valve.

5.1.1 Level A. Preservation and packaging of equipment for Level A shipment shall be in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with Process C-1 and drying, if required, shall be accomplished with Method D-1, or D-4. Preservation and packaging shall be Method IA.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging shall be accomplished in accordance with the supplier's own commercial practice when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C, as specified. See 6.2. Shipping container shall contain one unit package.

5.2.1 Level A. The equipment, when preserved and packaged in accordance with 5.1.1, shall be overpacked in containers complying with MIL-P-116, Method II D.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The equipment, when preserved and packaged in accordance with 5.1.3, shall be overpacked in compliance with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2.4 Cushioning or dunnage. Supplier shall provide sufficient cushioning or dunnage material to adequately protect the valve from damage due to shock and vibration loads encountered during all modes of handling and transportation. Shock and vibration tests shall be in accordance with this specification.

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5.3 Marking. In addition to the markings required by the contract order, units packages and shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The valve is intended for use as part of a torpedo fuel delivery system. It opens when gas pressure from a combustion chamber drives a piston-knife into a disc and ruptures it. The fuel then flows, from the fuel tank to the combustion chamber through the valve.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Assigned activity for preproduction tests
c. Any exceptions to this specification
d. Inspection data required
e. Required level of preservation, packaging and packing

6.3 Definitions

6.3.1 Ready. The condition of the valve when fully assembled, after completion of servicing and maintenance, with the ports plugged.

6.3.2 Operating. The condition of the valve when operating.

6.3.3 Non-operating. The condition of the valve when stored, transported, or handled. Tests on the valve, in the non-operating condition, shall be performed without packaging or packing.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration is gravitational units. The frequency is given in cycles per second.

6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 seconds. Acceleration is given in gravitational units.

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6.3.6 Shock . Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build-up to 90 percent of peak force in less than 0.010 seconds. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds and shock wave form Equivalent to one half sine wave. Duration is defined as the interval between the point of zero amplitude, from which the shock waveform builds up, to its decay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the valve is exposed, is expressed in degrees Fahrenheit.

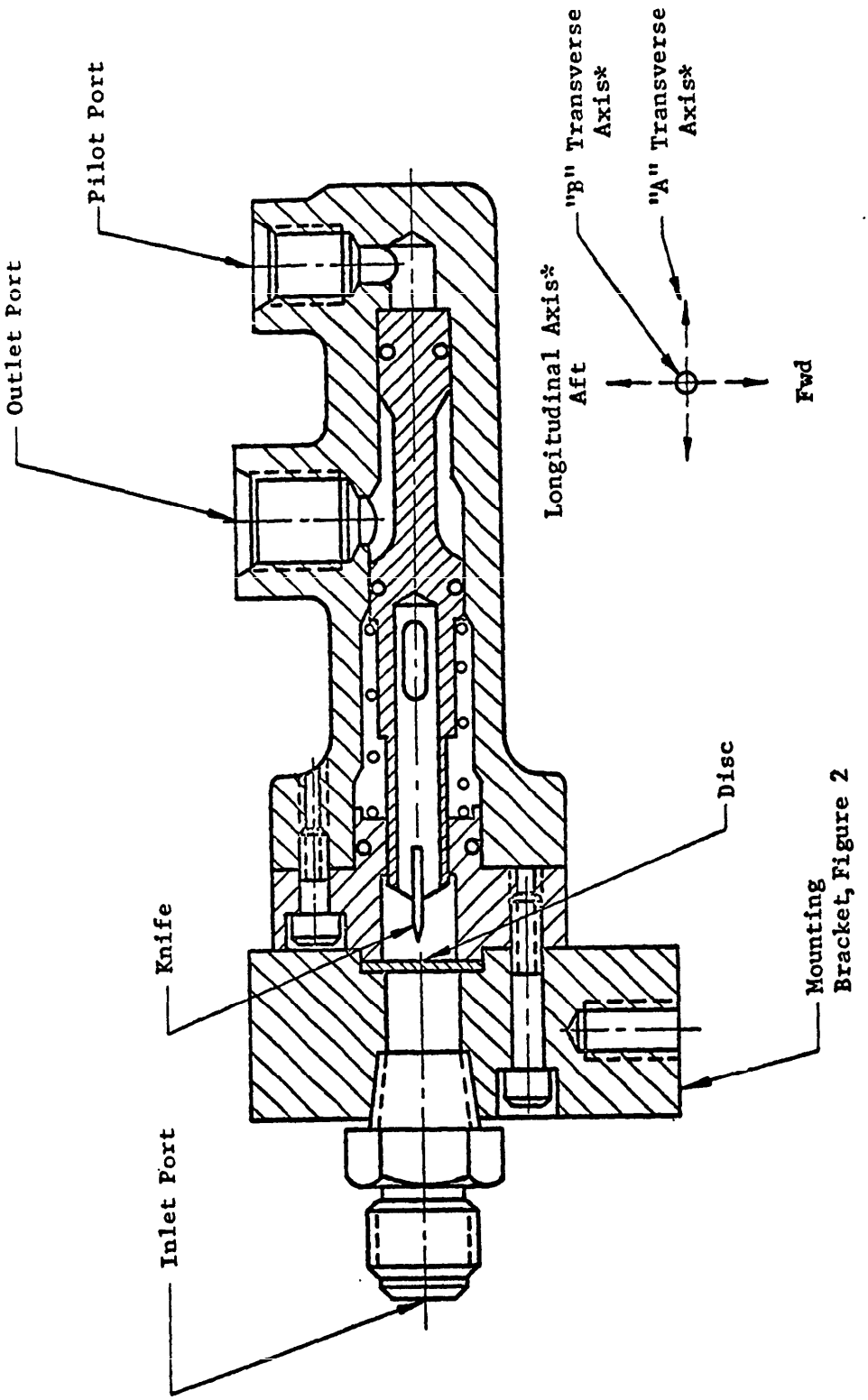
6.3.8 Humidity. Humidity is the ralative humidity of the environment to which the valve is exposed, expressed in percent.

6.4 Weight. The weight of the valve shall not exceed 1 pound.

6.5 Storage life.

6.5.1 Packed and Packaged condition. The valve shall be capable of withstanding storage of 5 years in the packed and packaged condition and, thereafter, meet the performance requirements of this specification.

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*The axes are designated in accordance with the operational mounting of the valve.

Figure 1, Operational Fixture

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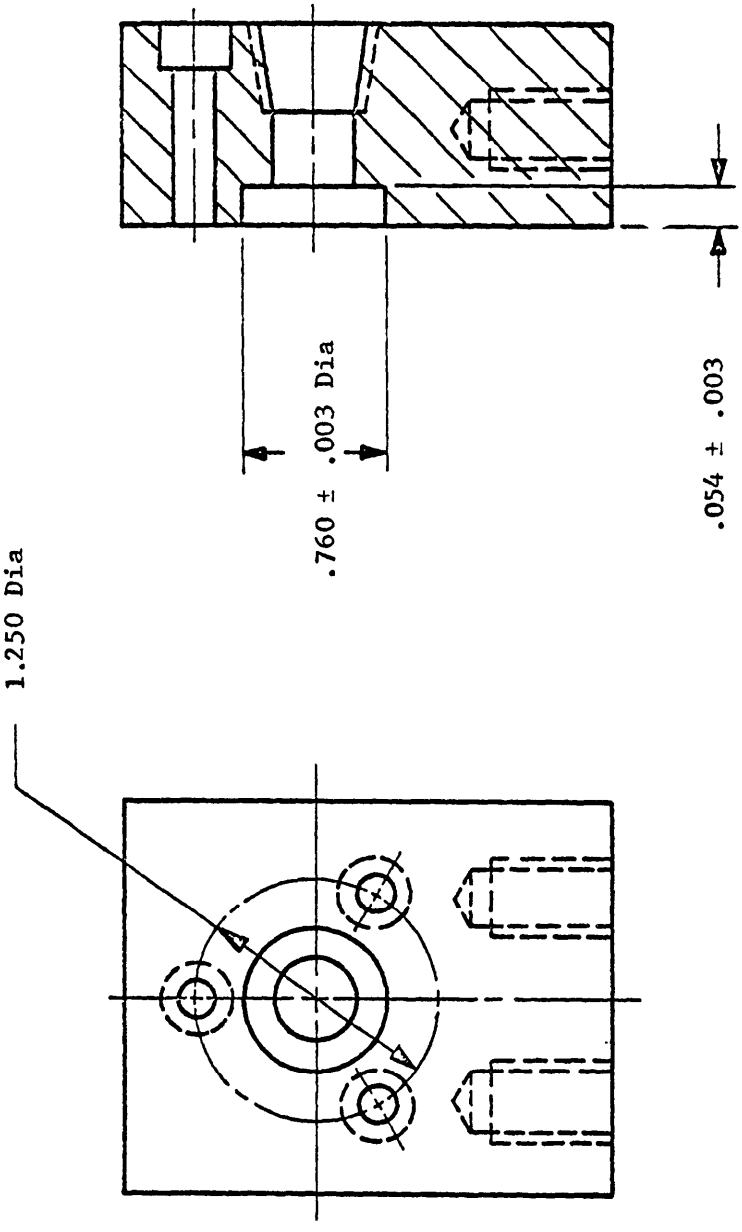


Figure 2, Mounting Bracket

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-V-82440(OS) VALVE, FUEL INTERLOCK

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

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